

Amine Benkia

Software Engineering Graduate · Full-Stack & Backend · AI-Augmented Development · End-to-End Ownership
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Summary

Software Engineering graduate from ENSEEIHT (INP Toulouse) with hands-on experience owning features **end-to-end in production SaaS** – from root-cause diagnosis of a data access bottleneck (**50% CPU reduction** on an enterprise pipeline) to design, rollout and live validation. Builds AI directly into the engineering workflow: shipped a **CI/CD-integrated LLM code reviewer** and a fully local RAG assistant. Comfortable across stacks: TypeScript/Node.js, Java/Spring Boot, Python, PostgreSQL, Docker/Kubernetes.

Work Experience

Padoa

Mar. 2026 – Present

Software Engineering Intern – TypeScript, Node.js, Angular, PostgreSQL, Elasticsearch, Docker

Paris, France

- Delivered a production feature end-to-end – REST API, business logic, Angular frontend, automated tests and progressive rollout via feature flags – from design to live validation.
- Diagnosed and eliminated a critical data access bottleneck on a high-throughput production pipeline, achieving a **50% reduction in CPU usage** across enterprise accounts including Airbus and Thales.
- Executed a backward-compatible PostgreSQL schema migration across multiple services: refactored SQL and Elasticsearch queries, enforced data integrity and ensured no regression across dependent services.
- Investigated and resolved production incidents by correlating logs, profiling slow queries and tracing cross-service call chains – identifying root causes and restoring system behavior without user impact.

CMI France

Jun. 2025 – Aug. 2025

Infrastructure & Automation Intern – Linux, Bash, PowerShell, Python

Paris, France

- Engineered automated monitoring workflows in Bash and Python, eliminating **~30%** of recurring manual checks and giving the team continuous visibility across the infrastructure.
- Audited 500+ user accounts and uncovered potential LOLBin attack surface in existing system configurations – produced a structured security report and escalated to the security team, directly informing hardening decisions.

Business Action Network

Jun. 2024 – Aug. 2024

Software Engineering Intern – Java, Spring Boot, PostgreSQL, REST APIs

Casablanca, Morocco

- Replaced a fully manual, email-based HR process with a Spring Boot backend (RBAC, electronic signatures, multi-step validation), cutting document turnaround from **1 day to a few hours** across 100 employees.
- Modeled the full document lifecycle in PostgreSQL (draft → review → signed → archived), implementing layered role enforcement across the REST API to enforce business rules at every stage.

Education

ENSEEIHT – INP Toulouse

2023 – 2026

Master's-level Engineering Degree in Computer Science

Toulouse, France

- Relevant coursework: algorithms, data structures, OOP, distributed systems, databases, cloud computing, networking, software design.

CPGE MPSI/MP

2020 – 2023

Classe Préparatoire aux Grandes Écoles – Mathematics & Computer Science

Casablanca, Morocco

Projects

AI Code Reviewer – GitHub Action

TypeScript · GitHub Actions · LLM · CI/CD

- Built a CI/CD-integrated code quality tool that triggers on every PR: statically analyzes diffs, invokes an LLM for bug detection and style review, and posts structured diagnostic reports as GitHub comments – embedding LLM review directly into the team's development workflow.
- Designed the pipeline as a composable TypeScript module – input normalization, LLM orchestration and structured output formatting – making it extensible to additional rule sets and linters without architectural changes.

MILO – Local-First AI Learning Companion

TypeScript · Chrome MV3 · Gemini Nano · RAG · pnpm monorepo

- Architected a fully local AI pipeline in a TypeScript pnpm monorepo: Chrome MV3 extension for content ingestion, vector-based indexing, RAG retrieval and on-device inference via Gemini Nano's Prompt API – no data ever leaves the device.
- Solved the core edge-inference challenge: managed context window constraints, built a retrieval fallback path and tuned chunking strategy to maintain response quality under on-device compute limits.

Technical Skills

Languages: Java, TypeScript, JavaScript, Python, SQL, Bash

AI-Augmented Engineering: RAG, LLM Applications, On-Device Inference, AI-Assisted Design/Testing/Documentation, Prompt Engineering

Backend: Spring Boot, Node.js, Express.js, REST APIs, Microservices

Data & Streaming: Apache Spark, Apache Kafka, Elasticsearch, PostgreSQL, SQL Optimization, Data Modeling

Cloud & DevOps: Docker, Kubernetes, Terraform, Linux, Git, CI/CD, Feature Flags

CS Foundations: Algorithms, Data Structures, OOP, Distributed Systems, Networking

Engineering Practices: Debugging, Performance Profiling, Testing, Code Review, Agile/Scrum

Languages

English: Professional (C1) | **French:** Fluent | **Arabic:** Native